San Francisco ATCT
Standard Operating Procedure
Version 1.11

List of Changes

VERSION	DATE	DESCRIPTION
1.0		
1.1	28FEB2019	Reformatted SOP for consistency purposes, combined NCT LOA into SOP.
1.2	20OCT2019	Removed VOX Channel due to AFV release
1.3	21MAY2020	Correct typos
1.4	03DEC2020	New diagrams in attachments, Added scratchpads, SVFR, LUAW procedures simplified, removed most taxiway restrictions and runway clearing for large aircraft, removed GH position and added TMC position, clarified VFR altitudes, updated taxiway Y references to taxiway Z2, added no s turn or 360s on final, missed approaches converted to table, changed Candlestick Point to Hunters point, Removed Military and Photo from local VFR, clarified TWR airspace is 2,000
1.5	07FEB2022	Readded dedicated Gate Hold position. LUAW clarification. Removed SFO_TMC position.
1.6	16JUN2022	Clean up unnecessary verbiage, re-format various tables, add route assignment table, update for new west end configuration, update missed approaches, add section about simultaneous departure procedures, add GC control for pushbacks on certain parts of the ramp, update diagrams, add ramp diagram, add VFR handoff boundaries, add runway distances from intersections, add equipment section and radar service notes
1.7	23FEB2023	Update simultaneous departure procedures, add detailed tower equipment info, fix clearance delivery frequency, change SNTNA to be preferred off 28s in 28/01, update noise abatement procedures, add new north field taxi procedures, update 28R/E assignment procedures, update missed approach instructions, update GAPP# 01 CFR to only be applicable to oceanic jets, correct A380/B748 taxi diagrams
1.8	30NOV2023	Update non-movement area pushback phraseology, update GC/LC jurisdiction diagrams, add scenery discrepancies attachment, fix SFOE initial altitudes, add holding between runways restrictions, clarify noise abatement section
1.9	11JUL2024	Updates for the new SEGUL# departures and removal of OFFSH#
1.10	07AUG2025	Explicitly allow NCT to clear into SFO Class B surface area, update west end phraseology, add ramp entry/exit procedures, add GC2 procedures, add LA position, clarify allowed departure headings, update diagrams, change gate hold procedures to be based on congestion instead of aircraft count
1.11	04SEP2025	Add note to use EFSTS procedures, update tower radar procedure description, add ramp gate assignment procedures

Section 1. General Information **5**
1-1 Purpose 5
1-2 Distribution 5
1-3 Cancellation 5
1-4 Equipment 5
1-5 Runway Use Waiver 5
1-6 Positions Table 6
1-7 Runway Configurations 6
1-8 Preferential Runway Assignment 6
1-9 Scratchpads 7
Section 2. Flight Data/Clearance Delivery **8**
2-1 General Procedures 8
2-2 IFR Departures 9
2-3 VFR Departures 10
2-4 Noise Abatement 11
Section 3. Ground Control **12**
3-1 Area of Jurisdiction 12
3-2 Separation 12
3-3 Runway Clearing and Crossings 13
3-4 Runway Assignment 13
3-5 Non-Movement/Movement Area Operating Procedures 15
Section 4. Local Control **16**
4-1 General Duties and Responsibilities 16
4-2 Area of Jurisdiction 16
4-3 Runway Selection 17
4-4 Position Procedures 17
4-5 Simultaneous Parallel Departures 18
4-6 Missed Approaches/ Go-Around Procedures 19
4-7 VFR Transition Procedures 20
4-8 Runway/Taxiway Restrictions 20
Section 5. Local Assist **21**
5-1 Position Procedures 21
Section 6. Gate Hold Procedures **21**
6-1 Position Procedures 21
Section 7. Split Ground Control Procedures **22**
7-1 Procedures 22
7-2 Jurisdiction 22
Attachment 1. A380 and B747-8 Taxi Diagrams **23**
Attachment 2. GC & LC Jurisdiction **25**
Attachment 3. SFO Gate Map **28**
Attachment 4. Ramp Diagram **29**

Attachment 5. Scenery Discrepancies **30**
Attachment 6. Runway Distance Remaining **32**
Attachment 7. Tower Delegated Airspace **33**
Attachment 8. VFR Handoff Boundaries **34**

Section 1. General Information

1-1 Purpose

This Standard Operating Procedure (SOP) outlines the procedures to be used by controllers working San Francisco ATCT positions on the VATSIM network, to ensure that traffic flows are handled in as efficient and timely a manner as possible. This SOP is for simulation purposes only and shall not be used for real world use or reference.

1-2 Distribution

This SOP is distributed to all members of the Oakland ARTCC on VATSIM.

1-3 Cancellation

All previous procedures are canceled.

1-4 Equipment

San Francisco International ATCT is equipped with a Certified Tower Radar Display (CTRD), ADS-B Airport Surface Surveillance Capability (ASSC), and Tower Data Link Services (TDLS) including its three parts, Flight Data Input/Output (FDIO), Digital Airport Terminal Information Services (D-ATIS), and Pre-Departure Clearance (PDC).

1-5 Runway Use Waiver

SFO has received a waiver to FAA Order 8400.9, National Safety and Operational Criteria for Runway Use Programs. Provisions of the waiver are:

a. Tower may operate on runways 28R/L and 01R/L with up to a 25-knot crosswind component on DRY runways, and up to a 10-knot tailwind component on runways "not clear or not dry". This waiver must NOT be used for runways 10/19.
   NOTE- *The term "Runways not clear or not dry" has replaced the term "wet runways".*
b. When using the provisions of this waiver on runways 01, the runways MUST be departure only.
c. Before using the provisions of this waiver on runways 28 or 01 Tower must check for FICON NOTAMs that indicate the runway condition. If the FICON shows no standing water or slippery surface conditions, tower may use the subject runways (28 or 01 only) under the provisions of the waiver.

1-6 Positions Table

The following position table details authorized positions for San Francisco Tower.

POSITION	RADIO CALLSIGN	FREQUENCY
Clearance Delivery	San Francisco Clearance	118.200
*Gate Hold**	*San Francisco Gate Hold*	*124.250*
Ground Control 1	San Francisco Ground	121.800
*Ground Control 2**	*San Francisco Ground*	*124.250*
Local Control	San Francisco Tower	120.500
*Local Assist**		
D-ATIS		118.850

** Indicates event only position, activated by the event CIC during published event times; or as practice in preparation for an event, up to 1 week prior to the event*

1-7 Runway Configurations

CONFIGURATION	DESCRIPTION
01/01	Landing and departing runways 01
28/01	Landing runways 28, departing runways 01
28 RT RT – Right Turn	Landing and departing runways 28 – TRUKN DP in use
28 SO SO – Straight Out	Landing and departing runways 28 – TRUKN DP not in use ***NOTE-*** *This is the preferred runway configuration for SFOW/OAKE*
10/10	Landing and departing runways 10
19/10	Landing runways 19, departing runways 10
19/19	Landing and departing runways 19

1-8 Preferential Runway Assignment

The table below shows the runway configurations in order of preference, most preferred at the top

Daytime (0600L – 0100L)		Nighttime (0100L – 0600L)	
West Plan	**Southeast Plan**	**West Plan**	**Southeast Plan**
28/01	19/10	28/28 RT	19/10
28/28	19/19	28/01	19/19
01/01	10/10	28/28 SO	10/10
		01/01	

1-9 Scratchpads

The following secondary scratchpads can be utilized by SFO ATCT or NCT to reduce verbal coordination.

ID	DEFINITION	ID	DEFINITION
2ER	Bay Tour	OPN	Operational Law Enforcement
BJJ	San Mateo Bridge	PIX	Photo Mission
CTN	Landing Seton Hospital	STF	Stanford University Hospital
GGB	Golden Gate Bridge	STK	Candlestick Point
HPT	Hunters Point	BNR	Banner Tow
LCL	Local traffic, no specific destination	USG	Benioff Children's Hospital
MPH	Mills Peninsula Hospital	XB	Aircraft issued a Class B clearance
LFT	Arrival to the left runway	RGT	Arrival to the right runway
VSL	Landing 28L visual separation with preceding aircraft	VSR	Landing 28R, visual separation with preceding aircraft

Section 2. Flight Data/Clearance Delivery

2-1 General Procedures

a. Issue departure clearance in accordance with current directives, Letters of Agreement and this section.
b. Issue TEC routes for aircraft with destinations within NCT (except RNO and satellites). If a pilot is unable to accept a TEC route, issue vectors direct destination and coordinate with NCT.
c. Issue Pre-Departure Clearances (PDC) for those aircraft so equipped.
d. When an aircraft requesting clearance requires route or traffic management coordination, advise the TMU/CIC so that they can complete the coordination prior to issuing the clearance.
e. Oceanic, Far East, and Cargo flights usually require runway 28L/R for departure for performance reasons. When runway 01 is the advertised departure runway, these aircraft may be issued runways 28L/R and assigned the appropriate SID.
f. Instruct aircraft that will push onto taxiway A, or onto the non-movement areas between boarding areas C and D, boarding areas D and E, and taxilane Y (depicted in Attachment 4), to contact Ground Control prior to pushback. All other aircraft will be instructed to report ready to taxi unless gate hold procedures are in effect.
g. When an arrival runway is closed at night, advertise the ILS to the parallel runway in the D-ATIS. Do not advertise visual approaches to the parallel. If there is no operable ILS approach to that runway, advertise the approach to that runway that has the most precision guidance laterally and vertically.
h. When gate hold procedures are initiated, complete a new D-ATIS advising that gate hold procedures are in effect.
i. If dual ground control frequencies are in use, assign aircraft to each ground control frequency as follows:
    i. GC1: Gates D9-G14, Signature Aviation, UAL Maintenance Operations Center, the superway, and all cargo aircraft not within an uncontrolled non-movement area
    ii. GC2: Gates A1-D8
j. When gate hold procedures are in effect, advise each aircraft to request push and start on the appropriate frequency.

    PHRASEOLOGY-

    READBACK CORRECT, CONTACT GROUND ON (frequency) WHEN READY FOR PUSHBACK

    READBACK CORRECT, ADVISE READY FOR PUSHBACK THIS FREQUENCY

2-2 IFR Departures

a. Standard DP/Route and Departure Sector

i. SFOW

DEST/ROUTE	RUNWAY	AIRCRAFT	DP	DEP SECTOR
Northbound	01 / *28**	P, T, J	TRUKN#	Richmond
	28		SNTNA#	
	01		SFO#	
	28		GAPP#	
Southbound	01	J	SSTIK#	Sutro
	28		WESLA#	
	01 / 28		SEGUL#	
		P, T, J	GAPP#	
Oceanic	28	J	GNNRR#	
			MOLEN#	
	01 / 28	P, T, J	GAPP#	

** SNTNA# shall be used instead of TRUKN# off Runways 28 while in 28/01 configuration*

ii. SFOE

DEST/ROUTE	RUNWAY	AIRCRAFT	DP	DEP SECTOR
Northbound	10 / 19	P, T, J	CIITY# GAPP#	Richmond
Southbound		J	SAHEY#	Sutro
		P, T, J	GAPP#	
Oceanic		J	MOLEN#	
		P, T, J	GAPP#	

b. Additional Non-DP Heading

i. Runways 01: 050°, 350°, and runway heading may be assigned after coordination with NCT

ii. Runways 10: 085° and runway heading may be assigned after coordination with NCT

c. Altitude Assignment
   i. SFO ATCT must issue interim altitudes as specified below when a Top of Climb (TOC) or Top Altitude is not published
   ii. SFOW

ROUTE	RUNWAY	AIRCRAFT	ALTITUDE
GAPP#, MOLEN#, SEGUL#, SFO#, WESLA#	28	P, T, J	3,000 or CVS x 3,000
All others	01 / 28	P, T	5,000 or CVS x 5,000
		J	10,000 or CVS x 10,000

   iii. SFOE

ROUTE	RUNWAY	AIRCRAFT	ALTITUDE
All	10 / 19	P, T	5,000 or CVS x 5,000
		J	15,000 or CVS x 15,000

   iv. When instructed by NCT to issue a non-standard interim altitude, advise the aircraft to expect filed altitude three (3) minutes after departure.

2-3 VFR Departures

a. Ensure VFR departures have their aircraft type, origin, and destination filled out in their flight plan.
b. Clear VFR departures out of the Bravo airspace. Do not clear the aircraft to a specific destination or fix.
c. Assign VFR Prop departures an interim altitude of (at or below) 2,000

   PHRASEOLOGY-
   CLEARED OUT OF SAN FRANCISCO BRAVO AIRSPACE, MAINTAIN VFR AT OR BELOW 2,000 WHILE IN BRAVO AIRSPACE. SQUAWK (code)

d. Assign VFR Jet departures an interim altitude of (at or below) 3,000
   i. When departing runway 28s, assign heading 280

   PHRASEOLOGY-
   CLEARED OUT OF SAN FRANCISCO CLASS BRAVO AIRSPACE, MAINTAIN VFR AT OR BELOW 3,000, DEPARTURE FREQUENCY (frequency), SQUAWK (code)

   CLEARED OUT OF SAN FRANCISCO CLASS BRAVO AIRSPACE, FLY RUNWAY HEADING, MAINTAIN VFR AT OR BELOW 3,000, DEPARTURE FREQUENCY 135.100, SQUAWK (code)

2-4 Noise Abatement

a. Departure noise abatement hours are daily from 2200L until 0700L Monday – Saturday and until 0800L on Sundays.
b. There are no preferred noise abatement procedures during SFOE.
c. SFO ATCT is in CFR status for all departures.
d. NCT may authorize a 050° heading for departures off runways 01 in lieu of the procedures outlined in the table below.
e. Noise Abatement DP/Route/Heading Assignments

DEST/ROUTE	RUNWAY	AIRCRAFT	DP/HDG	DEP SECTOR
Northbound	01 / 28	P, T, J	NIITE#	Richmond
Southbound (0100L-0500L)	01	P, T, J	NIITE# GOBBS	
All	01	P (non-RNAV)	RWY	

NOTE- *The procedures in this section do not cover all destinations for the entirety of noise abatement hours; standard procedures shall be utilized where not covered by this section*

Section 3. Ground Control

3-1 Area of Jurisdiction

a. Ground control has jurisdiction over all airport movement areas, **EXCEPT** those indicated below:
    i. Runway(s) designated active. The inactive runways shall be released to Ground Control.
    ii. All taxiways between active runways
    iii. When departing Runways 01 or 19, taxiways east of Alpha 1.
    iv. When landing Runways 01, all taxiways south of taxiway Mike 2.
    v. When departing Runways 19, all taxiways south of taxiway Mike 2.
    vi. The "West End"; Taxiway Zulu from Taxiway Charlie to Taxiway Zulu 2 as follows:
        1. When landing Runways 10 (SFO10);
        2. When departing Runways 28 (SFOW);
        3. When Local Control has been given jurisdiction.

PHRASEOLOGY-
ONCE AROUND THE WEST END / TWICE AROUND THE WEST END
WEST OF THE RIGHT BEHIND <CALLSIGN 1>, WEST OF THE LEFT BEHIND <CALLSIGN 2>
WEST END TO GROUND

3-2 Separation

a. Coordinate with LC for the use and/or crossing of an active runway and the West End. Report when runway/West End is clear.
b. Coordination between Local Control and Ground Control involving the opening/closing of active/inactive status of a runway shall be done through voice. Ground Control has jurisdiction over all inactive/closed runways and coordination with Local Control to use those runways is not required.
c. When Local Control has coordinated a helicopter landing or departing on the movement area, do not taxi any aircraft so that it would conflict with or be over-flown by the helicopter.
d. When in 28/01 configuration and taxiing FBO aircraft eastbound on taxiway C, GC must instruct the aircraft to hold short of taxiway E. After a correct readback, instruct the aircraft to monitor or contact LC as appropriate.
e. When issuing instructions for aircraft taxiing out of the FBO, GC must issue an initial turn instruction for taxiway C for the aircraft
    EXAMPLE-
    N451RY, runway 1R taxi via right turn C, Z, B1, A
    N451RY, runway 28R at E taxi via left turn C, hold short of E
f. When runways 01 are used for landing, ensure that all aircraft taxiing to Runway 01R are instructed to hold short of Mike 1.

3-3 Runway Clearing and Crossings

a. Be aware of the operational need for Local Control to issue appropriate turn instructions to aircraft exiting runways as necessary. Ensure that all runway exits are left unimpeded for arrival aircraft to taxi clear of the runway.
b. SFO has received approval to conduct multiple runway crossings in accordance with JO 7210.3. Approval authorized on the following taxi routes, all of which are less than 1000 feet between runway centerlines.
    i. Taxi routes crossing runways 28/10: Taxiways K, D, E, L, P, N, C, C3 to S1 to S to S3 to Z1, R, Runways 1R/19L, 1L/19R.
    ii. Taxi routes crossing runways 01/19: Taxiways E, V, C, F, F1, G, M, M1 to A2 to A1 to L2, Runways 28R/10L, 28L/10R
    iii. Multiple runway crossings are not authorized on taxiways H, Q or T.

3-4 Runway Assignment

a. Sequence taxiing aircraft to provide for minimal delay and an optimal order for expeditious takeoff clearance by Local Control. Aircraft subject to in-trail restrictions should be alternately sequenced with unrestricted departures.
b. Unless otherwise coordinated, departing aircraft shall be assigned runways as follows:
    i. Advertised Departing Runways 01:
        1. Right Turns – Runway 01R
        2. Left Turns – Runway 01L
        3. Oceanic, Far East, Cargo – Runways 28L/28R if operationally necessary
    ii. Advertising Departing & Landing Runways 28:
        1. Aircraft south of Runway 28L – Runway 28L
        2. Aircraft north of Runway 28R – Runway 28R
    iii. Advertising Departing Runways 10, Landing Runways 19:
        1. Aircraft south of Runway 10R
            a. Routed via CIITY# DP – Runway 10L/S1 via Z, S
            b. All other departures – Runway 10R/S3 via Z, S
        2. Aircraft north of Runway 10L – Runway 10L/C3
    iv. Advertising Departing & Landing Runways 10:
        1. Aircraft south of Runway 10R – Runway 10R/Z1
        2. Aircraft north of Runway 10L – Runway 10L
    v. Advertising Departing & Landing 19:
        1. Taxi all aircraft to Runway 19R. If an aircraft requests taxi to Runway 19L, issue instructions to Runway 19L and hold the aircraft short of the first active runway.
        2. If a runway other than advertised is requested, issue instructions to the requested runway with instructions to hold short of the first active runway.

c. During a L28/D01 operating, all departing aircraft parked on ramps north of runway 28R shall be routed to runways 01L or 01R around the west end of the airport via taxiway Z, except as follows:
    i. VFR aircraft shall be assigned 28R at taxiway E or 28R full length.
    ii. Lifeguard aircraft are exempt from this section.
    iii. Aircraft that identify an operational requirement for use of Runway 28R. (e.g. cargo aircraft parked at the northwest ramp)
    iv. Aircraft routed on a right-turn departure (TRUKN#, NIITE#) may be assigned runway 28R/E intersection departure after prior coordination with Local Control.
    v. Aircraft routed on a straight-out departure (everything except TRUKN# or NIITE#) may be assigned 28R/E without prior coordination as long as they are not routed over: OAK, V244, LIN, SAC, ORRCA, MOGEE, TIPRE, SYRAH, or landing the OAK Complex. Allowed routings include all departure fixes from DEDHD counterclockwise to NTELL.
    NOTE- *Discretion and good judgment must be applied in the decision to assign RWY 28R/E to ensure it benefits LC as well as the pilot. Runway 28R/E should not be assigned during times when operational benefit will not be realized by LC.*

3-5 Non-Movement/Movement Area Operating Procedures

a. All aircraft that push back onto taxiway Alpha shall be instructed "push back onto taxiway Alpha approved"
b. All aircraft that push back onto taxiway Alpha but need to be held short of Alpha shall be instructed "push back your discretion, hold short of taxiway Alpha". When aircraft can be pushed back onto Alpha, the aircraft shall be instructed "push back onto taxiway Alpha approved."
c. Aircraft that push back onto the non-movement areas between boarding areas C and D, boarding areas D and E, and taxilane Y (depicted in Attachment 4) will contact Ground Control for pushback. They shall be given appropriate pushback instructions and a spot to exit the ramp at (if applicable). For example, "push back your discretion, spot 5A" or “push back your discretion, taxilane Y.”
    i. Spots 5A & 5B, and spots 6A & 6B are usable by aircraft smaller than a B757 simultaneously. Aircraft that are a B757 or larger must use spots 5 and 6.
d. All other aircraft that do not push back onto the movement area shall be instructed "push back your discretion." Examples of the areas that do not push back onto the taxiway are cargo areas like Plot 41.
e. All inbound traffic shall be instructed to "taxi to the ramp." Specific instruction shall be given when operationally necessary.
f. Workload permitting, traffic advisories in the non-movement area will be given. Do not issue control instructions to aircraft in the non-movement area. Use only advisory phrases in exchange of traffic information.
g. When Ramp Control is operational, it may be assigned the responsibility for making gate assignments to all inbound aircraft, after coordinating with the event CIC and SFO ATCT. Ramp Control will directly communicate gate assignments to aircraft prior to their arrival.
h. Aircraft entering the A or G ramps shall be advised to enter via spots 1 or 11, with the following exceptions:
    i. ADG-VI aircraft (B748, A388) must enter via spot 10.
    ii. ADG-IV & ADG-V (e.g. B757, A330, B777) parking at B27 must enter via spot 2.

Section 4. Local Control

4-1 General Duties and Responsibilities

a. LC is responsible for runway separation and control, sequence, and separation of IFR, SVFR, and VFR aircraft in the San Francisco ATCT delegated airspace shown in Attachment 7.
    i. NCT retains the authority to issue Class B clearances for the portion of the surface area delegated to Local Control.
    ii. LC provides Class B radar service within their delegated airspace.
    iii. LC radar identifies all VFR non-jet departures and advises them of radar contact. LC makes handoffs for these departures to NCT.
    i. NCT sequences VFR high performance turboprop & jet arrivals.
b. Verbally coordinate:
    i. All departures from a non-advertised runway.
    ii. All Runway 28 RT departures during 28/01 and 28/28 SO configurations.
    iii. All departures when the assigned DP is not printed on the strip/ flight plan.
c. The following departures are subject to release by NCT:
    i. Runway 01 departures to Napa CX (KAPC, KSTS, KDVO, etc.)
    ii. Runway 01 GAPP# jet oceanic-bound departures
    iii. During SFOE, departures landing in the San Jose complex.
d. NCT has control for turns after a departure leaves 1,500 feet.
e. Verbally CFR all departures during noise abatement operations.
f. Utilize the Electronic Flight Strip Transfer System (EFSTS) as notification for each IFR and VFR jet departure.
    i. EFSTS must be utilized prior to verbal coordination or releases being initiated.

4-2 Area of Jurisdiction

a. Local Control has jurisdiction over the following airport surfaces:
    i. Runway(s) designated active. The inactive runways shall be released to Ground Control.
    ii. All taxiways between active runways.
    iii. When departing Runways 01 or 19, taxiways east taxiway Alpha 1.
    iv. When landing Runways 01, all taxiways south of taxiway Mike 2.
    v. When departing Runways 19, all taxiways south of taxiway Mike 2.
    vi. The "West End"; Taxiway Zulu from Taxiway Charlie to Taxiway Zulu 2 as follows:
        1. When landing Runways 10 (SFO10);
        2. When departing Runways 28 (SFOW);
        3. When Local Control has been given jurisdiction

PHRASEOLOGY-

ONCE AROUND THE WEST END / TWICE AROUND THE WEST END

WEST OF THE RIGHT BEHIND <CALLSIGN 1>, WEST OF THE LEFT BEHIND <CALLSIGN 2>

WEST END TO LOCAL

4-3 Runway Selection

a. Runway selection shall be made in accordance with Sections 1-5 and 1-8 of this SOP
b. During arrival noise abatement (0100L – 0600L), LC shall advertise Quiet Bridge Visual approaches if feasible, otherwise they shall advertise the Runway 28R ILS.

4-4 Position Procedures

a. Separate/space arrival, departure and transitioning traffic in accordance with JO 7110.65 and current directives. Local Control may transfer communications and control of departing aircraft maintaining visual separation to NCT.
b. LC shall provide Class B radar service within their delegated airspace.
    i. This may involve radar identifying VFR aircraft (and advising them of radar contact) and making radar handoffs to NCT.
    ii. LC shall quick look the NCT position(s) working SFO finals. NCT shall not make radar handoffs for IFR arrivals to LC (VFR arrivals/overflights can be handed off).
c. Coordinate with GC, through the local assist when staffed, for the use and/or crossing of an active runway and the West End. Coordination between Ground Control and Local Control shall be conducted via voice. Report when the runway/West End is clear.
d. Coordination between Local Control and Ground Control involving the opening/closing or active/inactive status of a runway shall be recorded. Ground Control has jurisdiction over all inactive/closed runways and coordination with Local Control to use those runways is not required.
e. Coordinate with Ground Control a helicopter landing or departing on the movement area under Ground Control's jurisdiction.
f. Provide instructions to VFR non-jet departures so they may safely exit Bravo airspace.
    EXAMPLE-
    "Make right crosswind departure to follow highway 101" / "Make left turnout to intercept the coastline" / "Make a left turn to pass over the tower heading 050"
g. Issue taxi instructions to aircraft exiting the runways when compliance will be required prior to changing that aircraft to Ground Control frequency. Inform Ground Control when an aircraft is instructed to turn onto a taxiway that is Ground Control's jurisdiction.
h. When there are two grounds and an aircraft that is parking in GC2's jurisdiction exits runways 28 at a point west of taxiway D, Local Control will instruct that aircraft to contact GC1.
i. S-turns or a 360-degree turns are not authorized on the final approach course. Aircraft requesting these procedures shall be given missed approach instructions for re-sequencing.

j. Line up and wait (LUAW) procedures shall be applied as prescribed in JO 7110.65 and 7210.3. For the purposes of simulation, it can be assumed that SFO has a full operating safety logic system. Due to special authorizations from the FAA SFO may do the following differently:
    i. LUAW is authorized at the following intersections between sunset and sunrise:
        1. 1R/M
        2. 1L/M, 1L/H
        3. 10L/S2, 10L/R
        4. 10R/S3, 10R/R
    ii. Only one aircraft may be held in position on a given runway at a time. (e.g. cannot have 1 aircraft in position at 10L/S2 and another at 10L/S1 at the same time) between sunset and sunrise.
        1. Aircraft may be held in position simultaneously on runway 1L and 1R (e.g. 1 aircraft on 1L while another is on 1R).

4-5 Simultaneous Parallel Departures

a. When utilizing simultaneous departure procedures, LC shall ensure aircraft turn onto their assigned heading or toward their assigned RNAV fix prior to transfer of communications.
b. When both aircraft are flying an RNAV SID, simultaneous departures shall be conducted in accordance with FAA JO 7110.65 5-8-3 (c).
    i. SFO parallels are spaced by 750 feet so a 1000 foot stagger shall be utilized when applying these procedures.
    ii. This procedure is not applicable when the following airplane requires wake turbulence separation with the leading airplane.
    iii. Tower applied visual separation may be applied to waive the stagger requirement provided it is maintained until aircraft achieve 1 mile separation in accordance with FAA JO 7110.65 5-8-3 (a).
c. When one or both of the aircraft are on a conventional departure procedure, simultaneous departures shall be spaced by 1 mile separation, in accordance with FAA JO 7110.65 5-8-3 (a).
    i. Tower applied visual separation may be applied provided it is maintained until aircraft achieve 1 mile separation.
d. Simultaneous departures from Runways 10
    i. Verbally coordinate auto-releases on heading 085° and runway heading with NCT
        1. NCT will place OAK into call for release status
    ii. Verbally coordinate termination of headings
    iii. Runway 10L: Richmond departures only
        1. Issue a heading of 085°
        2. Aircraft cannot be a super, heavy, A321, B717, or B38M
    iv. Runway 10R: Sutro departures only
        1. Issue runway heading
        2. Aircraft can be any category
    v. Aircraft departing into the same sector may be approved with verbal coordination; ensure no crossovers

4-6 Missed Approaches/ Go-Around Procedures

a. All go-arounds and missed approaches must be coordinated with NCT. Coordination shall include route and altitude, which may vary from those listed below as required for separation. Transfer of communication shall occur within 1/2-mile of the departure end of the runway or as soon as possible after conflicts are resolved.

b. Tower must issue the following to all unplanned instrument missed approaches:

RUNWAY	INSTRUCTION	AUTHORIZED HEADINGS	ALTITUDE	SECTOR / FREQ
01L / 01R	As coordinated	345° to 125° CW	As coordinated	As coordinated
10L / 10R	As coordinated	345° to 125° CW	As coordinated	Sutro – 135.100
19L / 19R	Published missed	None	3,100	
28L / 28R	Published missed	None	3,000	

c. When needed for separation between a departure and a missed approach, or a visual approach go-around, local control may issue the following headings without pre-approval from NCT:

RUNWAY	HEADING	ALTITUDE	SECTOR / FREQ
19L / 19R	As coordinated (for visual approaches only)		Sutro – 135.100
	170°	3,100	
	RWY[1]*	*4,000**	
28L / 28R	RWY	3,000	
	260°	3,100	
	315°[2]*	*3,000*	

*[1] *Runway heading is authorized for aircraft on a 1.2 nm final or greater at or above 500 feet*

*[2] *Runway heading and heading 260° are preferred over heading 315°*

4-7 VFR Transition Procedures

a. Apply these procedures to VFR aircraft in the San Francisco Class B surface area
b. When in west plan, the following procedures shall be used to provide additional separation between runways 28 departures and VFR transitions in the runways 28 departure corridor
    i. When a runway 28R/L departure taxies across runways 1 or a runway 28R/E departure has been approved, LC shall clear the runways 28 departure corridor. The corridor is defined as that airspace from the approach ends of runways 1 clockwise to abeam Fullers Point, including overhead of SFO.
    ii. The Local controller can either hold the runways 28 departures or hold the transitions.
    iii. For transition aircraft that are north of SFO proceeding southbound:
        1. If the aircraft is north of Hunters Point, provide instructions to keep the airplane north of Hunters Point until the runways 28 departures are at least 2 miles west of SFO.
        2. If the aircraft is south of Hunters Point hold the departure until the aircraft is abeam the approach ends of runways 1.
    ii. For transition aircraft that are south of SFO proceeding northbound:
        1. If the aircraft is east/southeast of SFO, provide instructions to keep the aircraft east/southeast of the approach end of runways 1 until the runways 28 departures are at least 2 miles west of SFO.
        2. If the aircraft is west of the approach end of runways 1, hold the runways 28 departure until the transition aircraft is north of a point abeam Fullers point.
b. When there are projected periods of multiple runways 28 departures anticipated, or when advertising departing runways 28, the CIC shall coordinate with NCT and SQL to terminate VFR transitions in SFO airspace.
c. LC can request at any time to terminate Class B transitions due to traffic volume or complexity.

4-8 Runway/Taxiway Restrictions

a. A380, DC10, MD11 and L1011 aircraft are prohibited from departing runway 1L.
b. Aircraft will not be on taxiway L directly behind runway 1R when a runway 1R departure begins takeoff roll at full length.
c. No 4-engine or Super aircraft on taxiway A1.
d. Restrictions apply for holding aircraft between runways when the runway behind them is used for arrival or departure operations. When these restrictions are in effect, no arrivals shall cross the landing threshold and no departures shall begin takeoff roll on the runway behind:
    i. B748 or A380 holding between runways at any taxiway
    ii. A346 or B77W holding between runways except on taxiways H, T, or Q
    iii. A35K holding between runways on taxiway E or taxiway F1

Section 5. Local Assist

5-1 Position Procedures

a. Local assist performs duties outlined in JO 7110.65 2-10-3 (c) (2).

Section 6. Gate Hold Procedures

6-1 Position Procedures

a. Gate Hold may be operated when activated during an event period only. If Gate Hold is not opened as a separate position, it shall be operated by CD or GC.
b. Gate hold may not be operated as a separate position at the same time as ground control is split and its responsibilities shall be delegated to CD, GC1, or GC2.
c. In a L28/D01 configuration, it is preferable to open GC2 over gate hold.
d. Initiate Gate Hold Procedures when:
    i. Directed by the TMU/CIC/EC (in that order).
    ii. Aircraft awaiting departure are expected to cause congestion in the movement area, at the discretion of the controller. Generally, impacts are seen when:
        1. The line of waiting aircraft extends past taxiway G on B (runway 1L) or taxilane M2 on A (runway 1R) when in a L28/D01 or L01/D01 runway configuration
        2. The line of waiting aircraft extends past taxiway L on F/C when in a L28/D28 RT/SO runway configuration.
        3. The line of waiting aircraft extends past taxiway Z1 or C3 on Z when in a L19/D10 runway configuration.
        4. The line of waiting aircraft extends past taxiway B1 on B (runway 10R) or Z/R on C (runway 10L) when in a L10/D10 configuration.
        5. The line of waiting aircraft extends past taxiway C on E when in a L19/D19 configuration.
e. Notify all tower cab positions when Gate Hold procedures are initiated.
f. Advise aircraft that "gate hold procedures are in effect" when needed.

    PHRASEOLOGY-
    GATEHOLD PROCEDURES ARE IN EFFECT. CONTACT (position name) ON (frequency) WHEN READY FOR ENGINE START OR PUSHBACK

g. Gate Hold shall give special handling to EDCT and CFR aircraft.
h. When aircraft request engine start or push back, note the time they made their request, ensure that the aircraft has the current D-ATIS, request the gate number, if appropriate, and advise the pilot of the estimated delay in minutes.
i. Advise the pilot to monitor the frequency for updates.

j. Monitor the rate of arrivals and departures to ensure a proper number of aircraft are ready for departure; normally, no fewer than 5 and no more than 10 aircraft should be at the departure runway(s).
k. Adjust order and timing of aircraft release based on ground congestion and length of departure queues. Release aircraft to provide operation benefit for outbound flow, by being aware of varying destinations/exit fixes/DPs.
l. If the proper number of aircraft are at the runway(s), normally one aircraft should be released to Ground Control each time an aircraft departs, unless arrival traffic is light
    i. Avoid releasing an aircraft to Ground Control if its ramp area is congested.
    ii. When releasing an aircraft to Ground Control, Gate Hold should use phraseology that minimizes congestion on the Ground Control frequency.

    PHRASEOLOGY-
    (Callsign), MONITOR GROUND ON 121.800.

Section 7. Split Ground Control Procedures

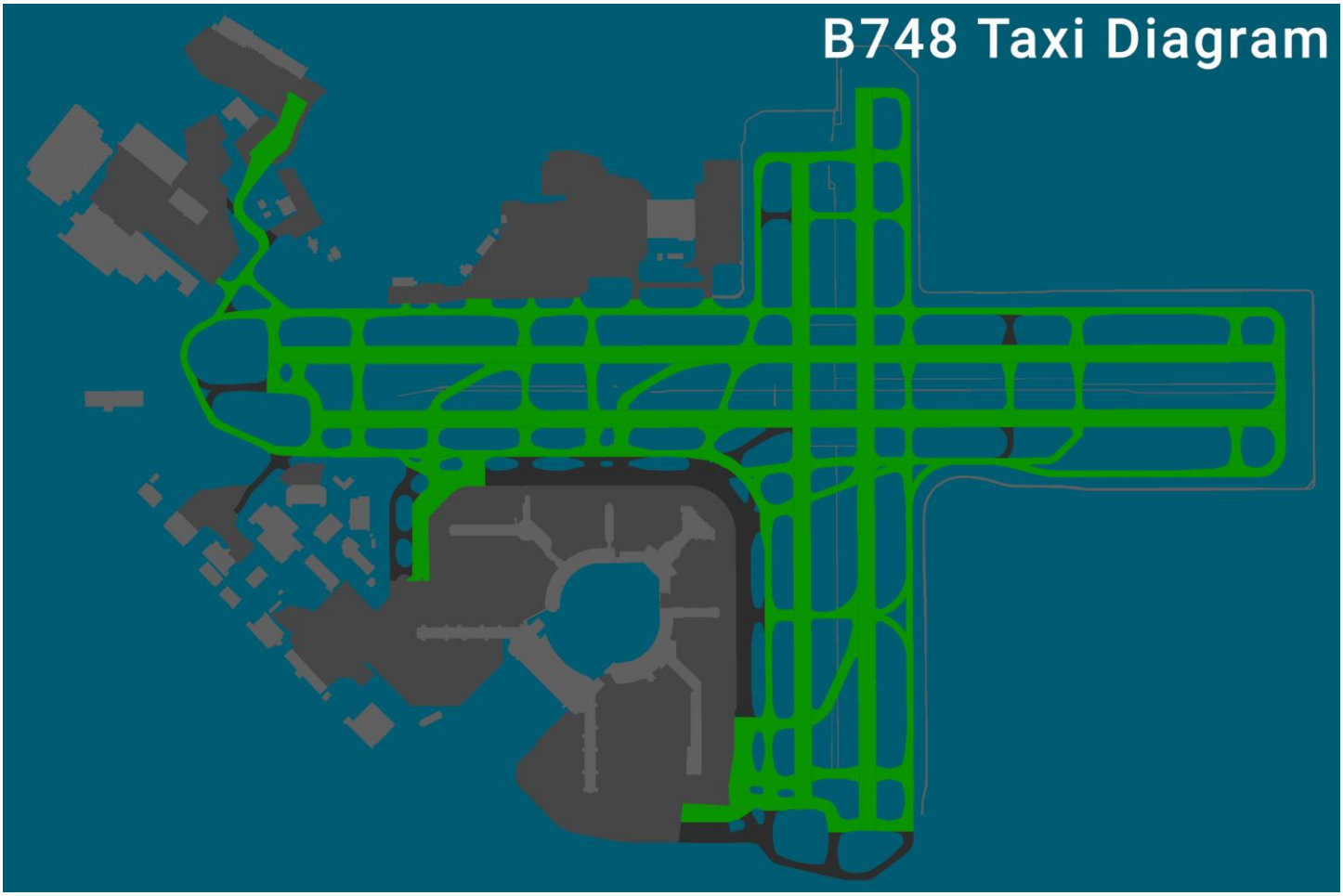
7-1 Procedures

a. When it is deemed appropriate during an event period, Ground Control may be split in accordance with Section 6-2. It is important that the two Ground Controls work well as a team and coordinate their work when needed.
b. Ground control must not be split in configurations other than 28/01.
c. Gate hold may not be operated as a separate position at the same time as ground control is split and its responsibilities shall be delegated to CD, GC1, or GC2.
d. When there are two grounds and an aircraft that is parking in GC2's jurisdiction exits runways 28 at a point west of taxiway D, Local Control will instruct that aircraft to contact GC1.

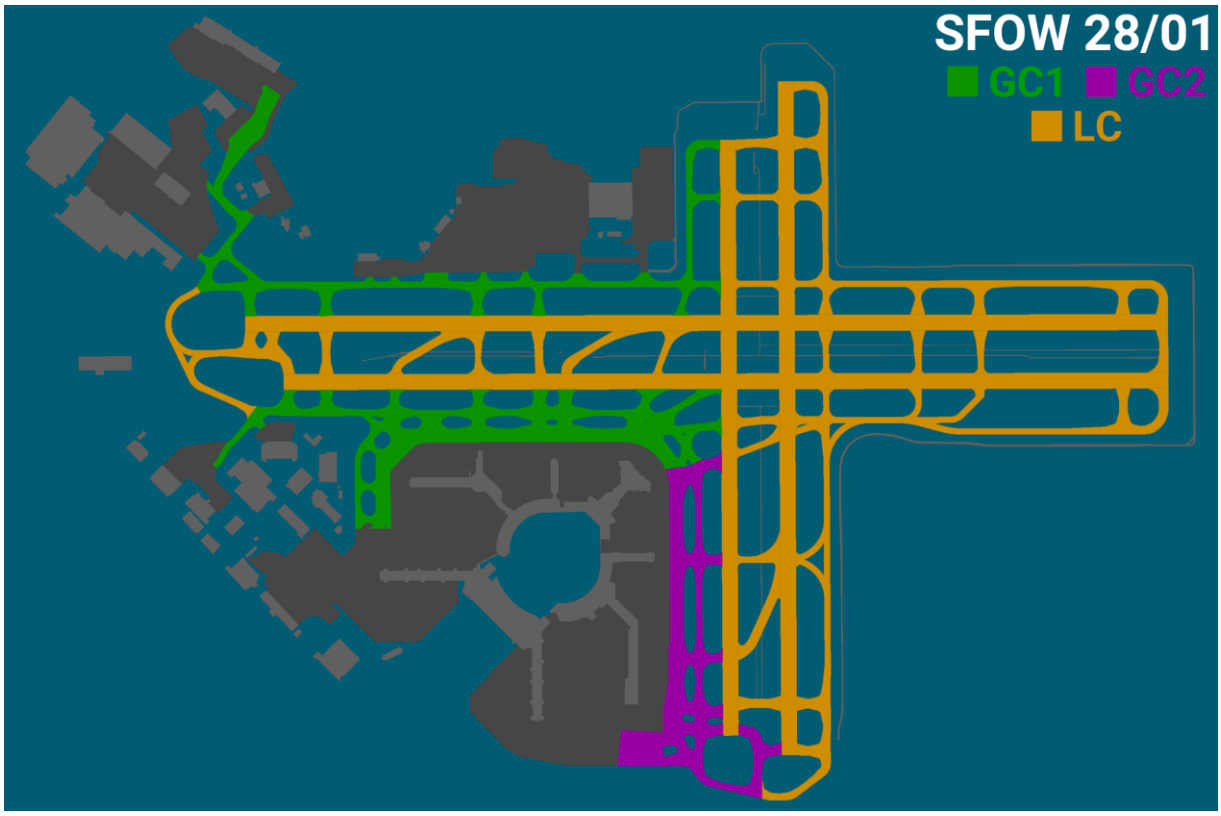
7-2 Jurisdiction

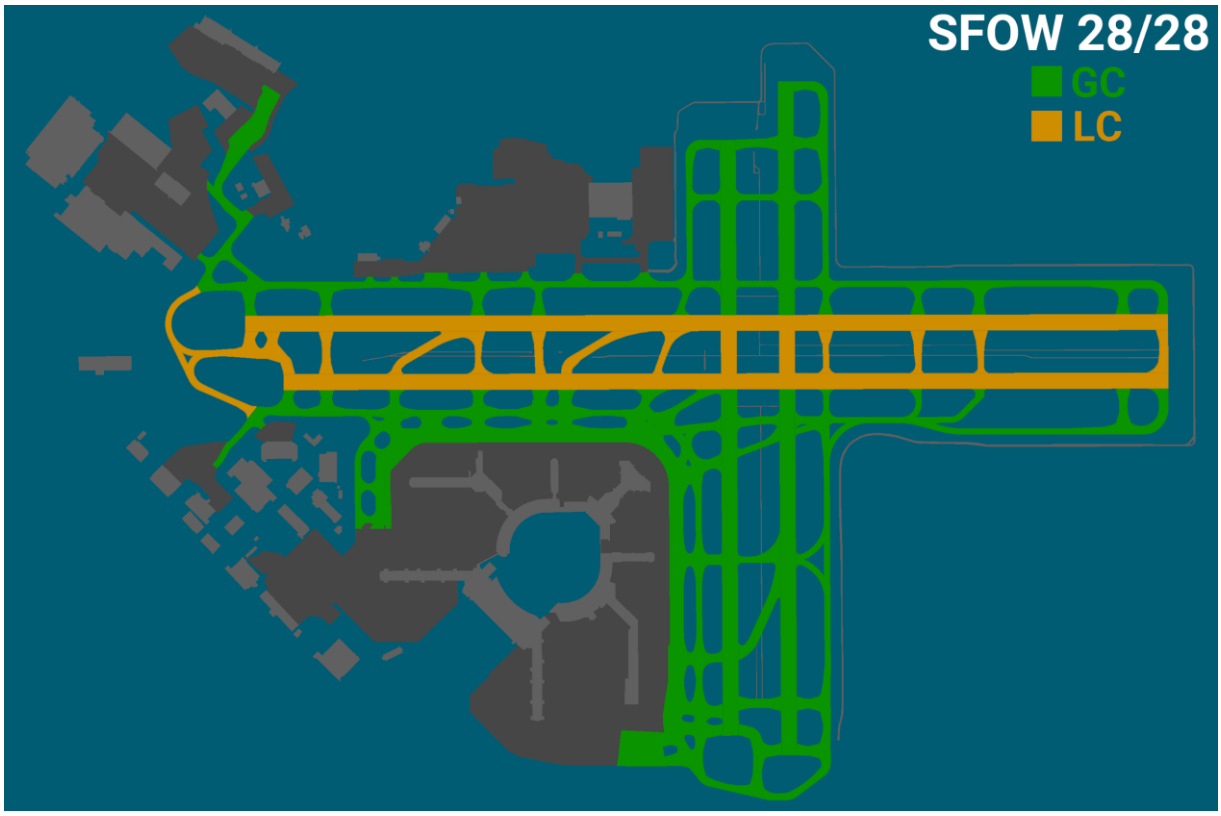
a. Ground Control 1 shall have jurisdiction over all movement areas controlled by Ground Control north of Gate D8, not including Gate D8 or taxiway F1.
b. Ground Control 2 shall have jurisdiction over all movement areas controlled by Ground Control from Gate D8 south, including taxiway F1.

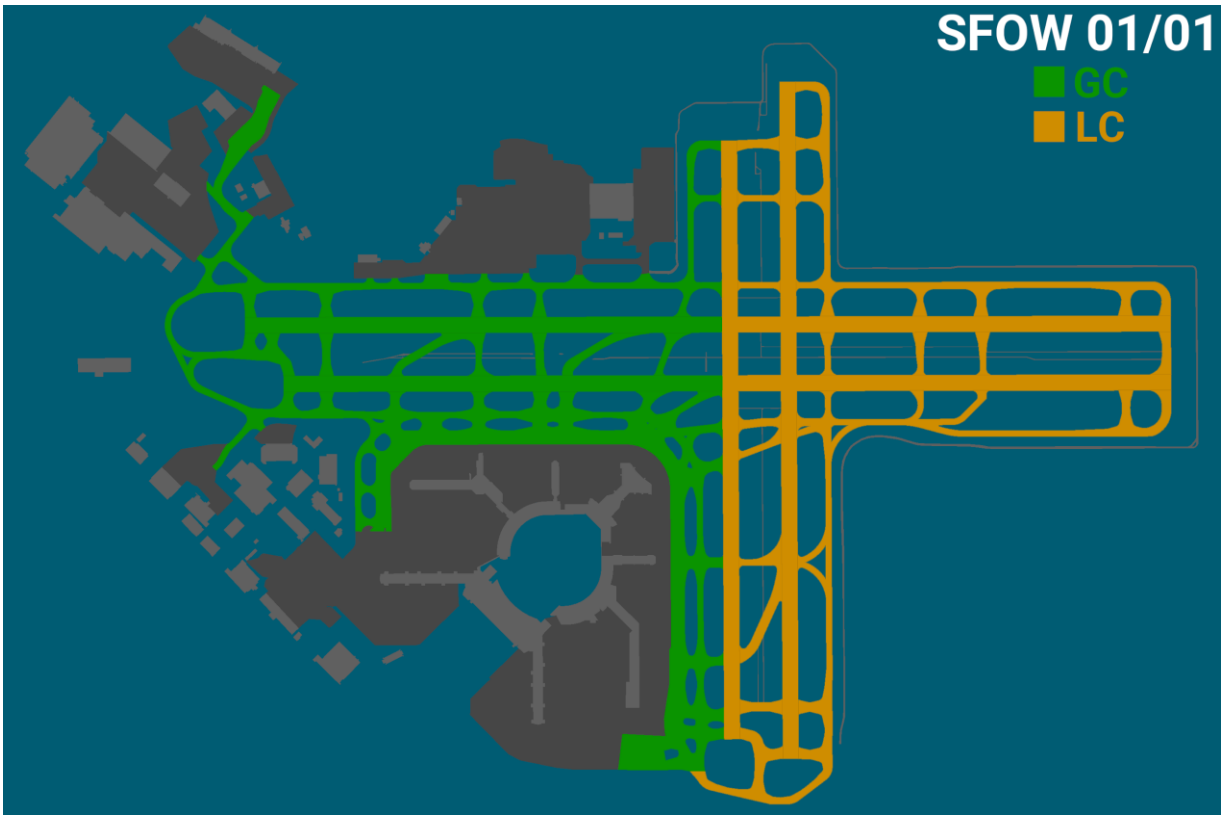
Attachment 1. A380 and B747-8 Taxi Diagrams

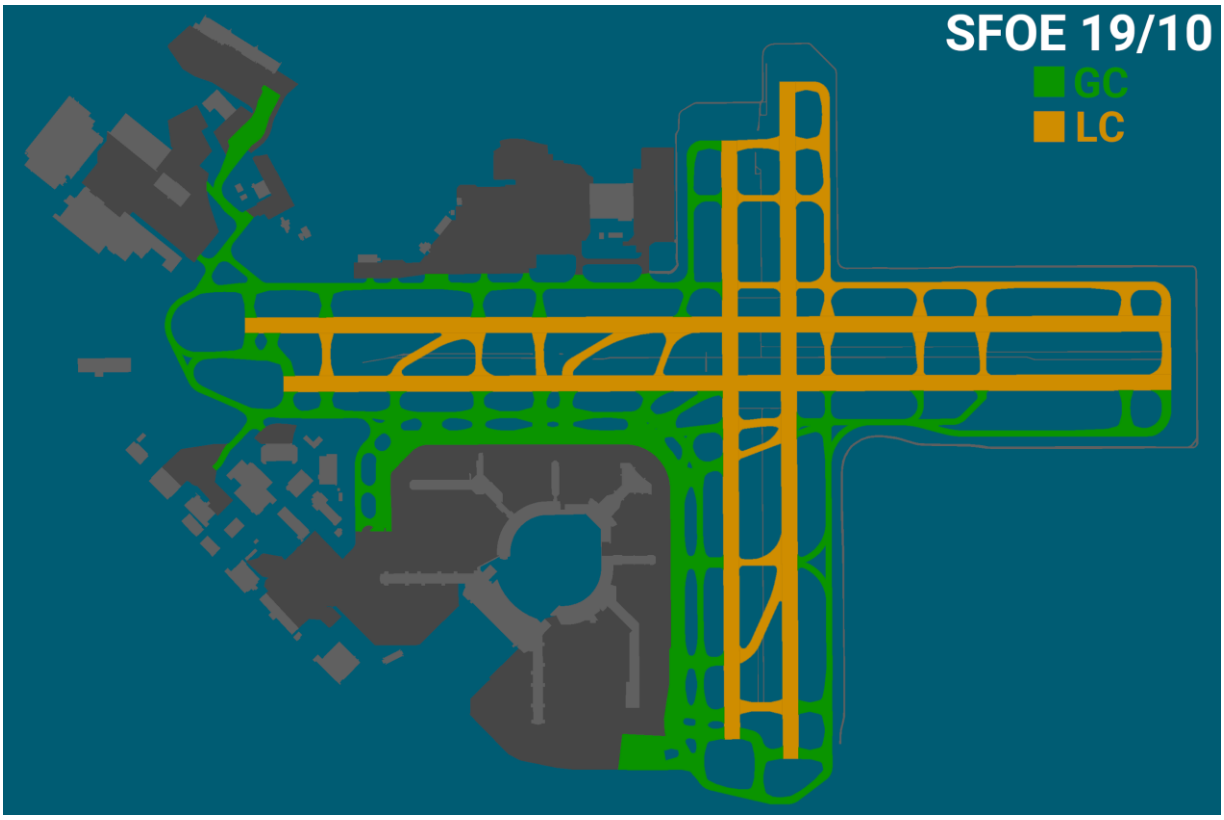

B748 Taxi Diagram

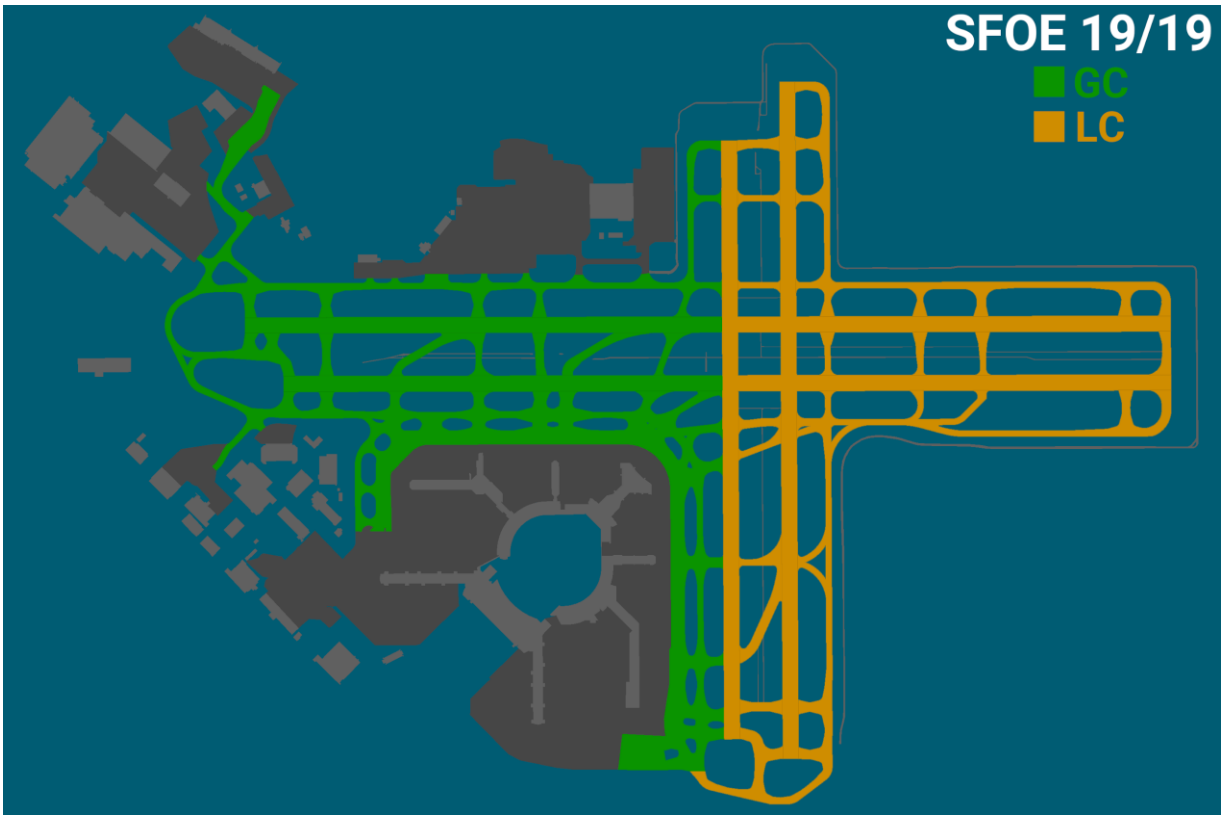
Attachment 2. GC & LC Jurisdiction

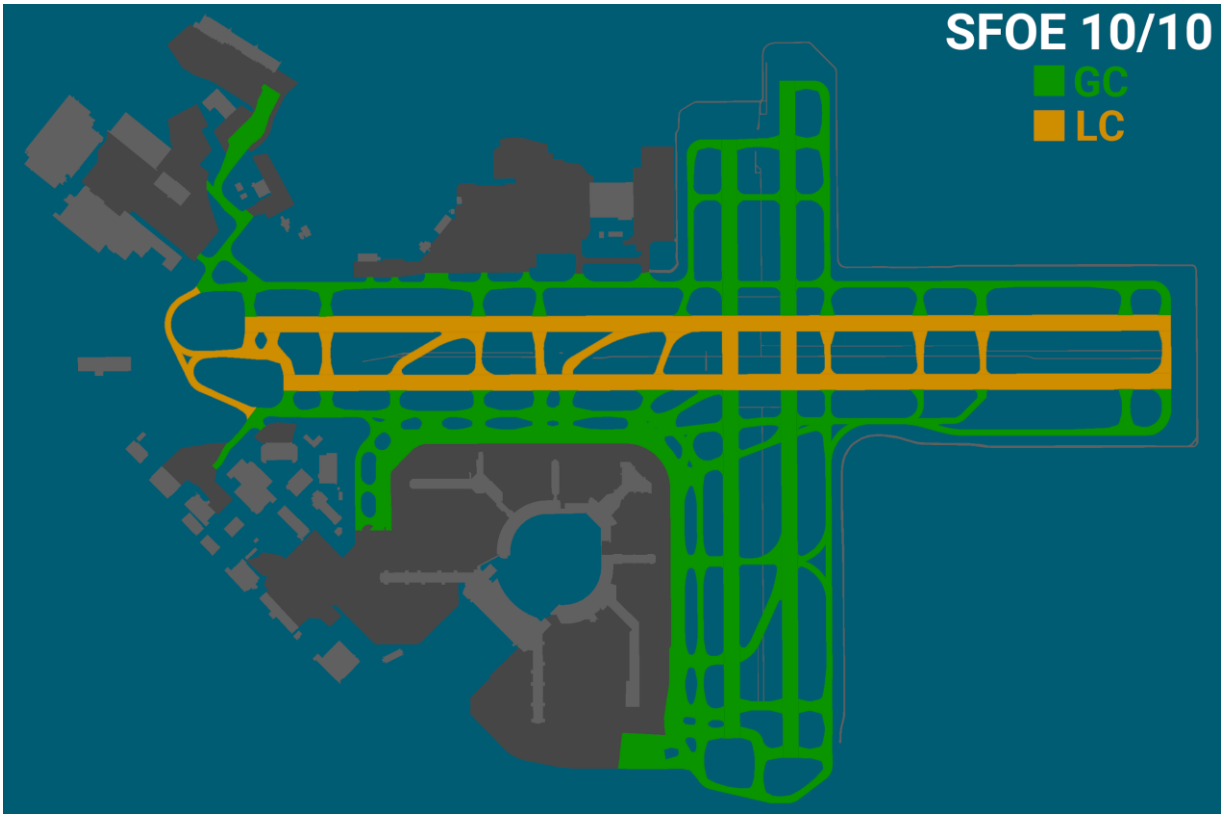




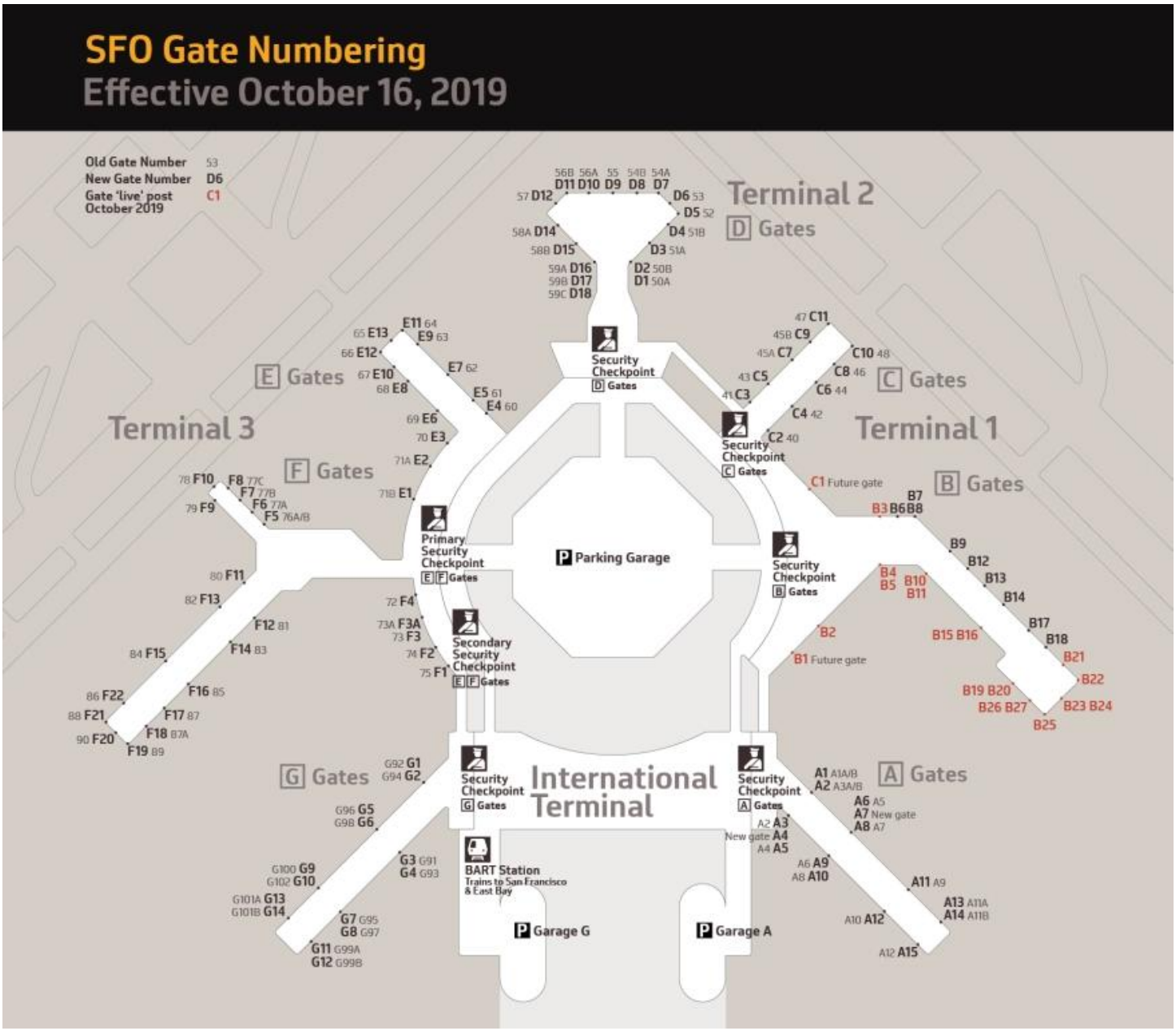

SFOW 01/01
GC
LC


SFOE 19/10
GC
LC

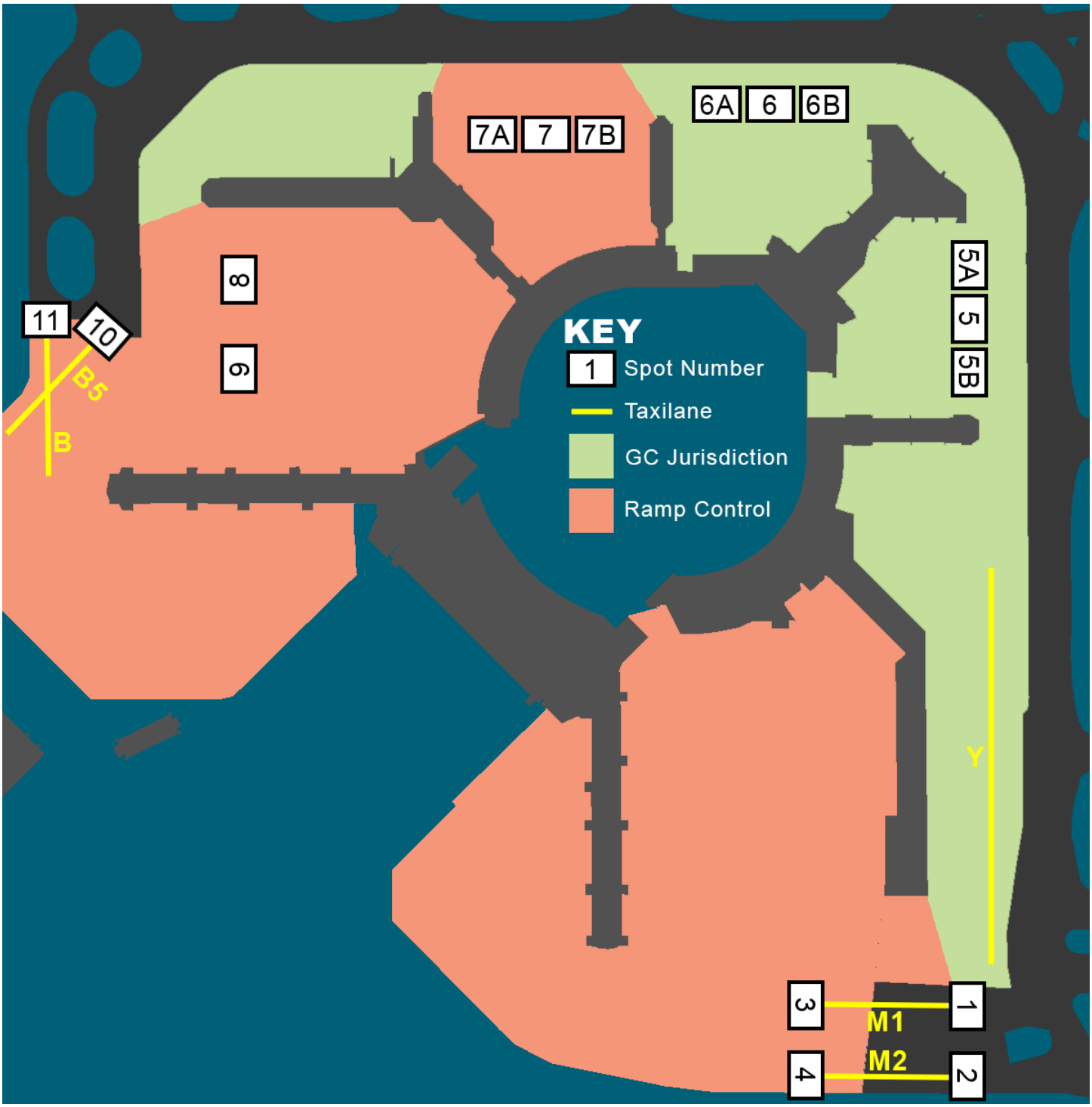

SFOE 19/19
GC
LC


SFOE 10/10
GC
LC

Attachment 3. SFO Gate Map

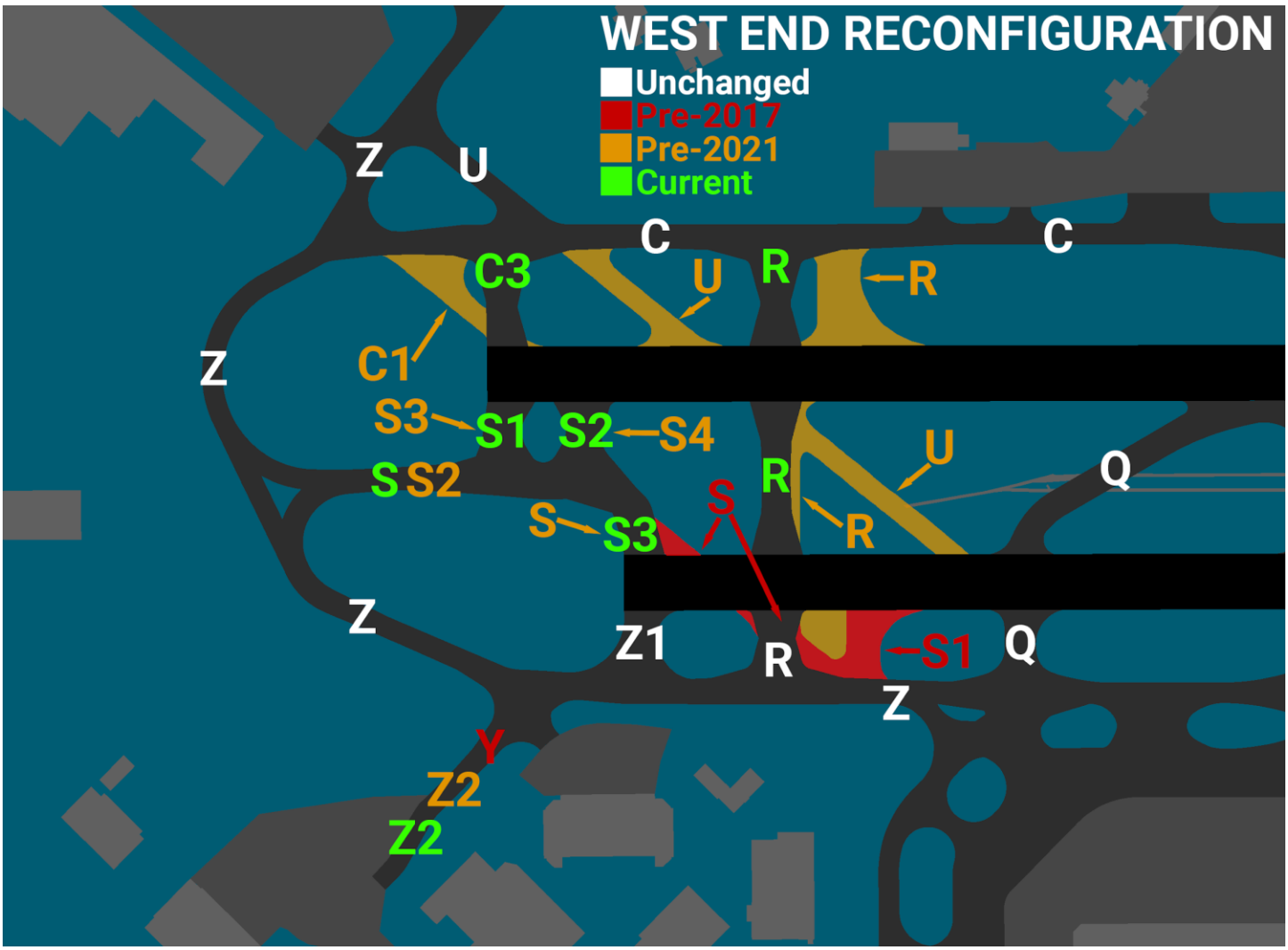


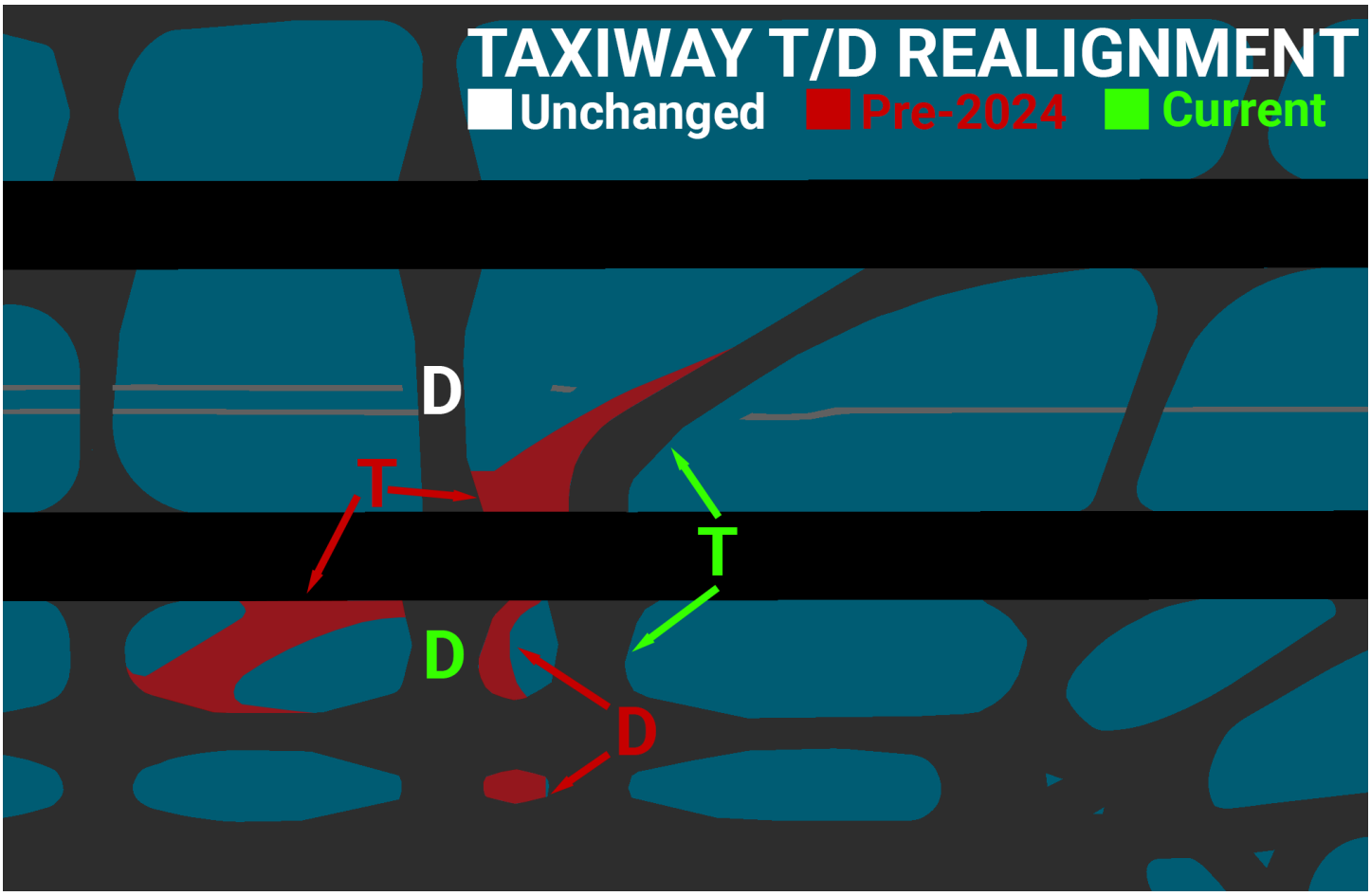
Attachment 4. Ramp Diagram



Attachment 5. Scenery Discrepancies

- The B gates were rebuilt 2017-2019
- Taxiway C2 was designated W prior to 2021
- Taxiway F2 was built in 2017
- The West End was reconfigured in 2017 and 2021 - see diagram below
- Taxiways T and D were realigned in 2024 - see diagram on next page




TAXIWAY T/D REALIGNMENT
Unchanged
Pre-2024
Current
D
T
T
D
D

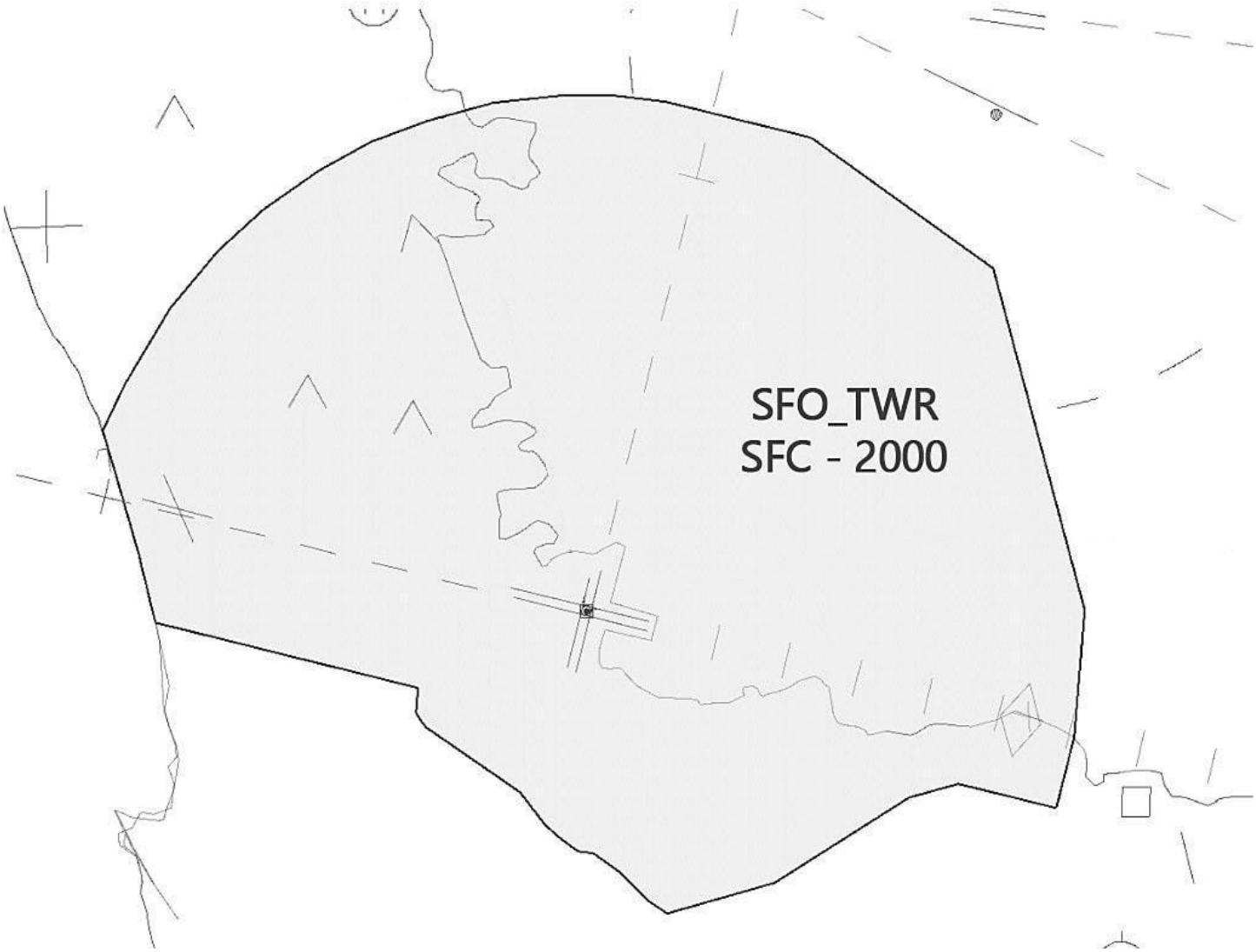
Attachment 6. Runway Distance Remaining

Note: full length distances not shown

TWY/RWY	DIST. 28L	DIST. 28R	TWY/RWY	DIST. 10L	DIST. 10R
C2	N/A	11,250	S2	11,200	N/A
F2	10,800	N/A	R	10,300	10,300
N	8,950	9,450	Q	N/A	9,650
P	8,150	8,600	K	8,550	8,550
L	6,950	7,450	D	7,750	7,750
01R/19L	6,450	6,950	T	N/A	
01L/19R	5,700	6,200	E	6,000	6,250
E	4,800	5,550	01L/19R	5,300	5,300
T	N/A		01R/19L	4,550	4,550
D	3,250	3,750	L	4,050	4,050
K	2,500	2,950	P	2,900	2,900
Q	1,400	N/A	N	2,100	2,100
R	750	1,200	F2	N/A	250
S2	N/A	350	C2	250	N/A

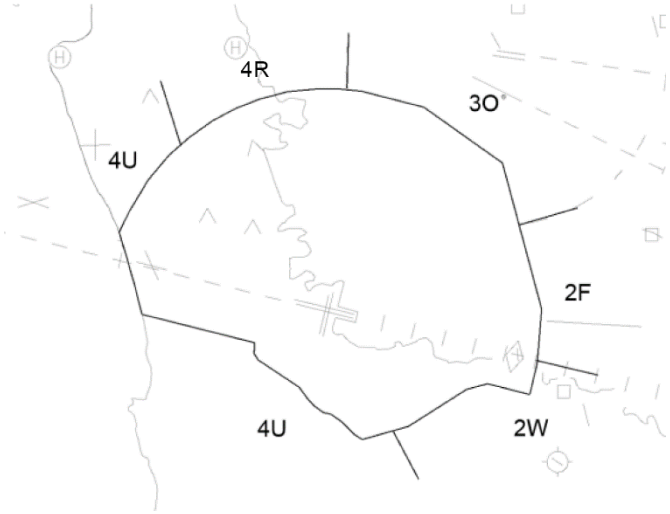
TWY/RWY	DIST. 01L	DIST. 01R	TWY/RWY	DIST. 19L	DIST. 19R
M (west)	7,300	N/A	E	7,850	FULL
M (east)	7,200	7,950	V	7,150	6,900
H	6,600	N/A	C	6,050	5,800
G	5,350	6,100	28R/10L	5,550	5,300
F1	N/A		28L/10R	4,800	4,550
F	3,650	4,350	F	4,300	4,000
28L/10R	3,100	3,850	F1	N/A	
28R/10L	2,350	3,100	G	2,500	2,250
C	1,850	2,600	H	N/A	1,000
V	700	1,450	M (east)	650	400
E	N/A	750	M (west)	N/A	300

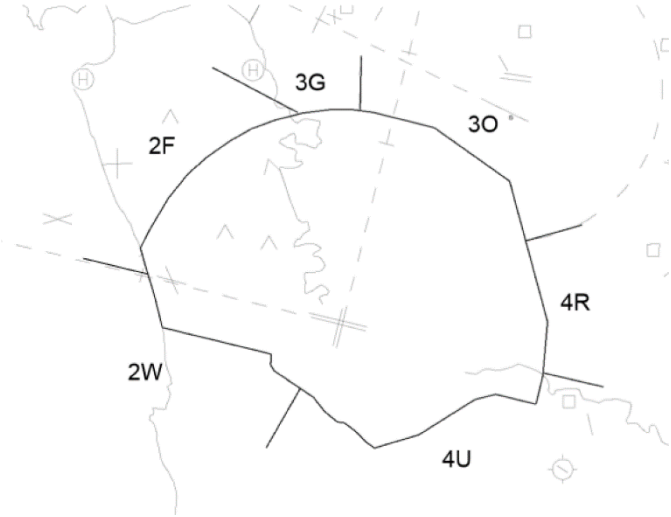
Attachment 7. Tower Delegated Airspace



Attachment 8. VFR Handoff Boundaries

a. SFOW



b. SFOE (10/19)

c. SFOE (10/10)